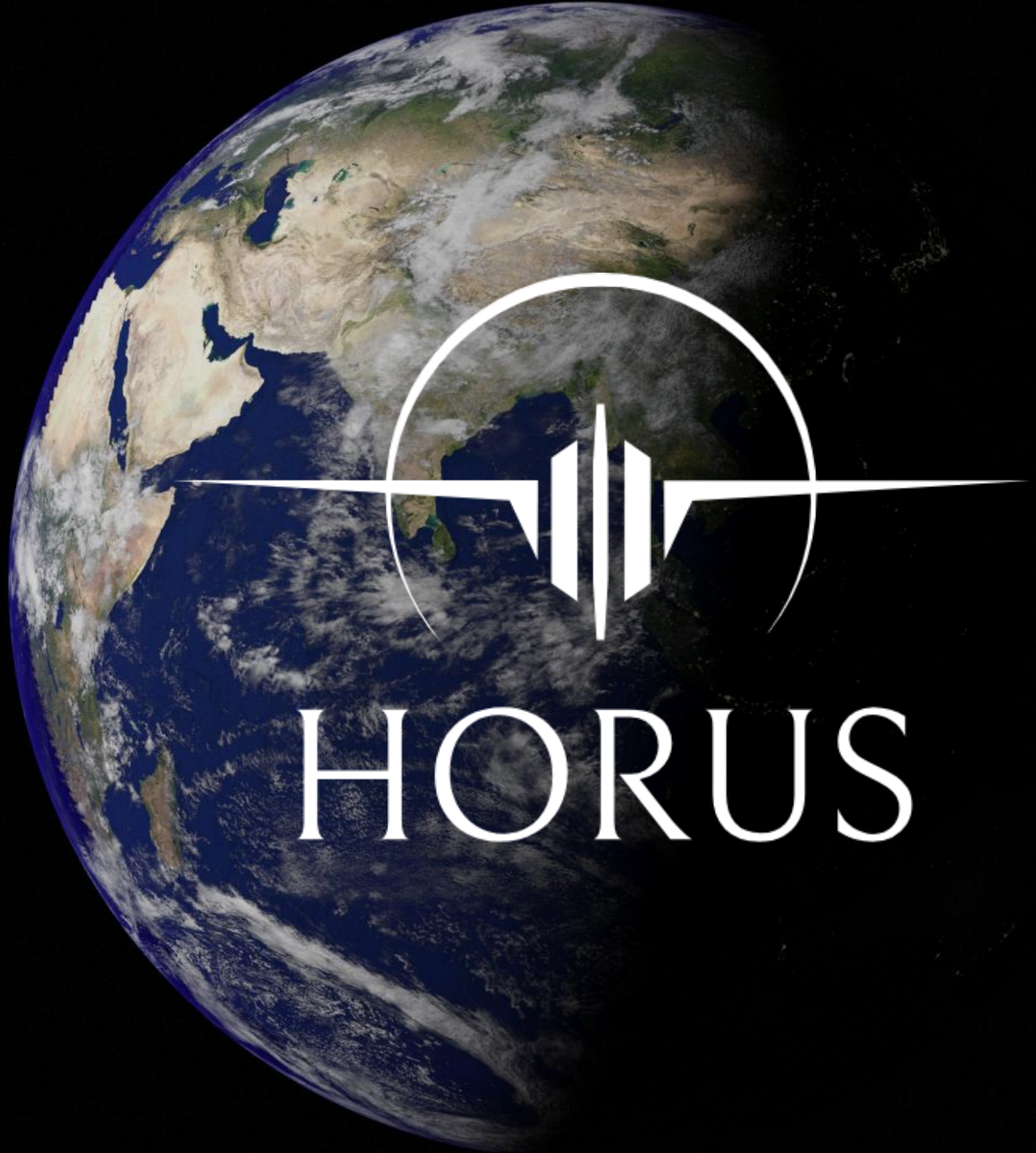
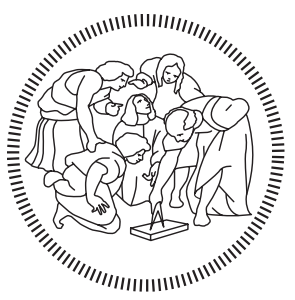




POLITECNICO
MILANO 1863



HORUS



POLITECNICO
MILANO 1863



HORUS MISSION

Final Presentation

Applied Space Mission Analysis and Design

2024/2025

Prof. Michelle Lavagna

Bellini D. , Beretta B., Biraghi G., Braga M., Cappuccio M., Cataffo F., Cesarini L., Dowell M., Juvara M., Pacursa A.J.

School of Industrial and Information Engineering
Master of Science – Space Engineering

ATOX and N₂ degradation

Criticalities

- Ionization and **damaging** of surfaces
- Surface **charging**

High atmospheric density

Criticalities

- Rapid orbit **decay**
- Increased **fuel consumption** for SK
- Higher aerodynamic torque

UV radiation

Criticalities

- Variation in surface **optical properties**

Charged Particles

Criticalities

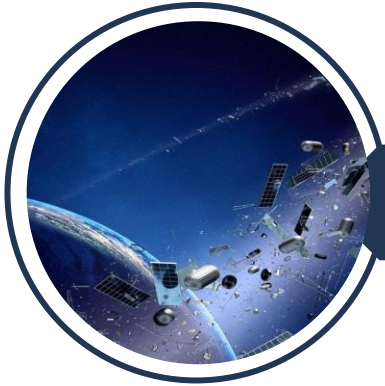
- Possibility of **SEE**
- **Radiation damage**

Frequent Eclipses

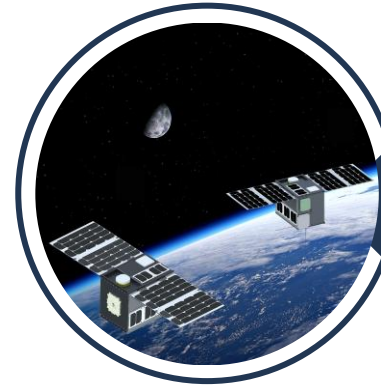
Criticalities

- **Variable heat flux** extremes

Why VLEO?



Reduced space debris for a cleaner orbital environment



Cost-effective high-resolution imaging



Enhanced telecommunications & improved revisit times



New frontier for Earth Observation

High-resolution



To provide specific **Earth coverage** areas with **high spatial resolution**, below **1 m**

Orbiting



To ensure **flexibility** in terms of target covered

Reconnaissance



The mission shall provide a **time to fly** not larger than **1 – 2 months**, and have an **operational life** as far as the attentioned event lasts

HORUS

Universal



To **limit** the **time to launch** and launch **costs**

Surveillance



Data shall be **downloaded** within **1 h after acquisition**

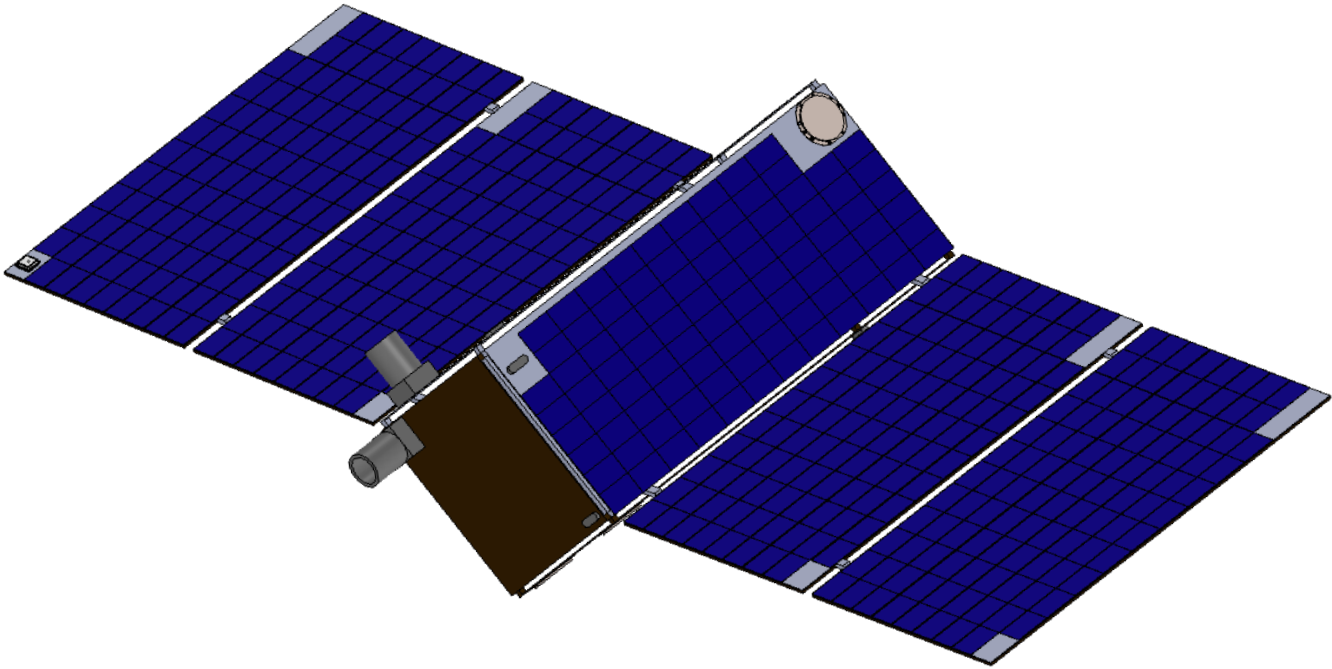


Technology and Launcher shall be **European**. To demonstrate the **TRL for VLEO**

Single drone

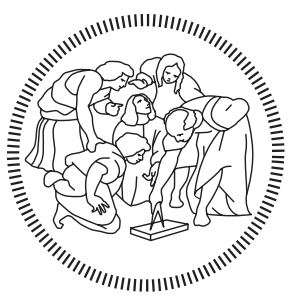
2x Optical Payload

Electric Propulsion



Wet Mass [kg]	Height [mm]	Width [mm]	Lenght [mm]
104.67	830	330	330

Mass and Size of a single drone



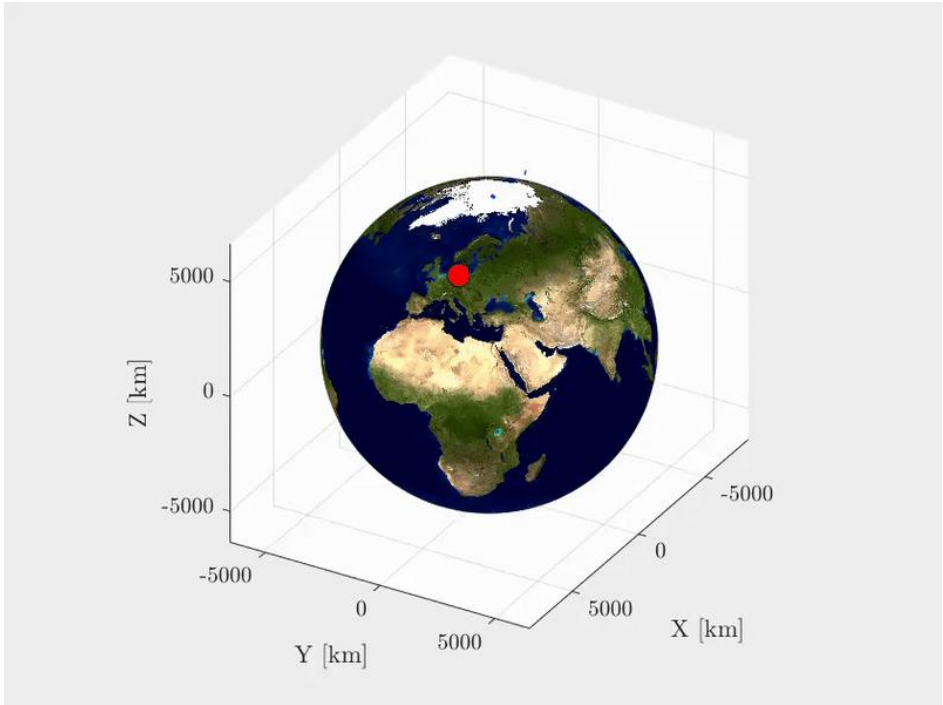
POLITECNICO
MILANO 1863



Mission Analysis

School of Industrial and Information Engineering
Master of Science – Space Engineering

Altitude [km]	Eccentricity [-]	Inclination [deg]	RAAN [deg]	AOP [deg]
262.57	0	90	Function of target longitude and starting epoch	Function of target longitude (time invariant)



Challenges

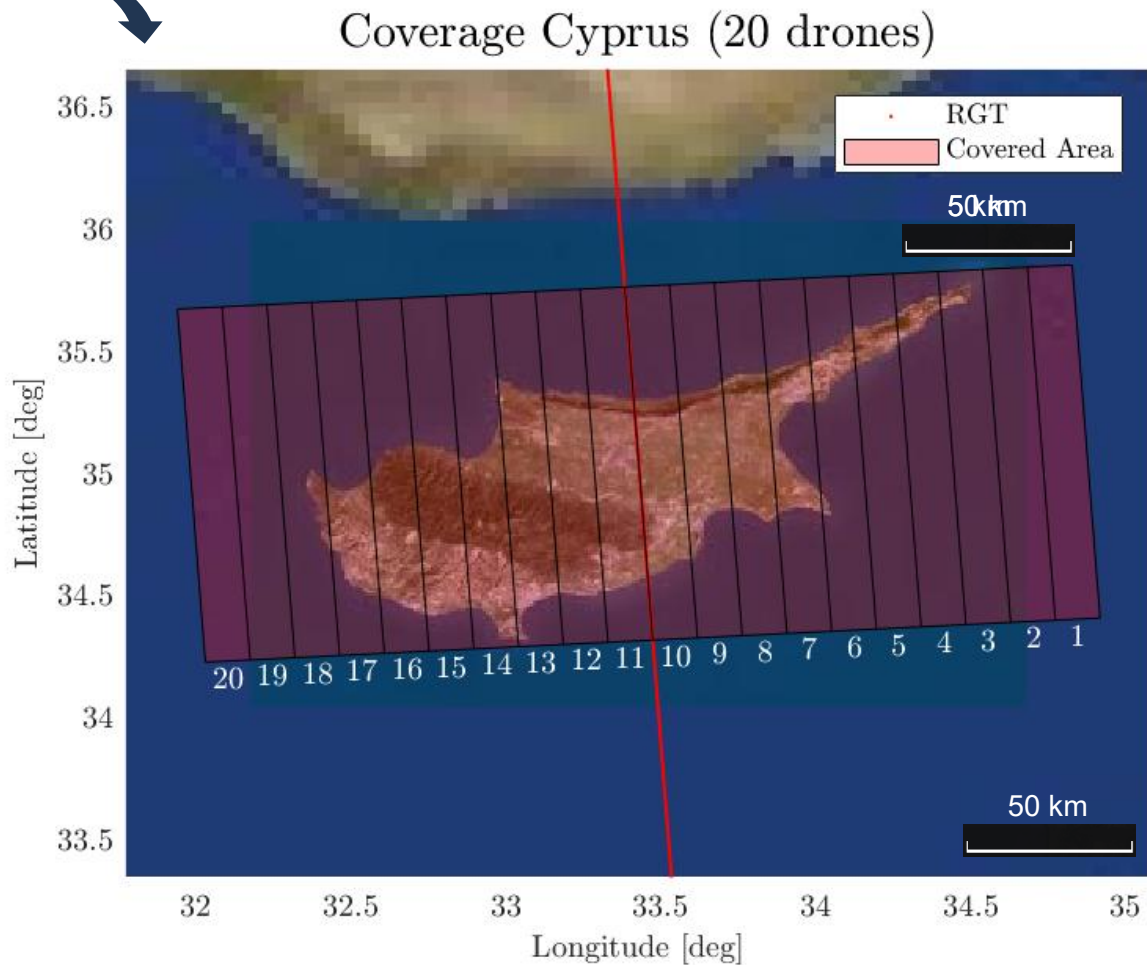
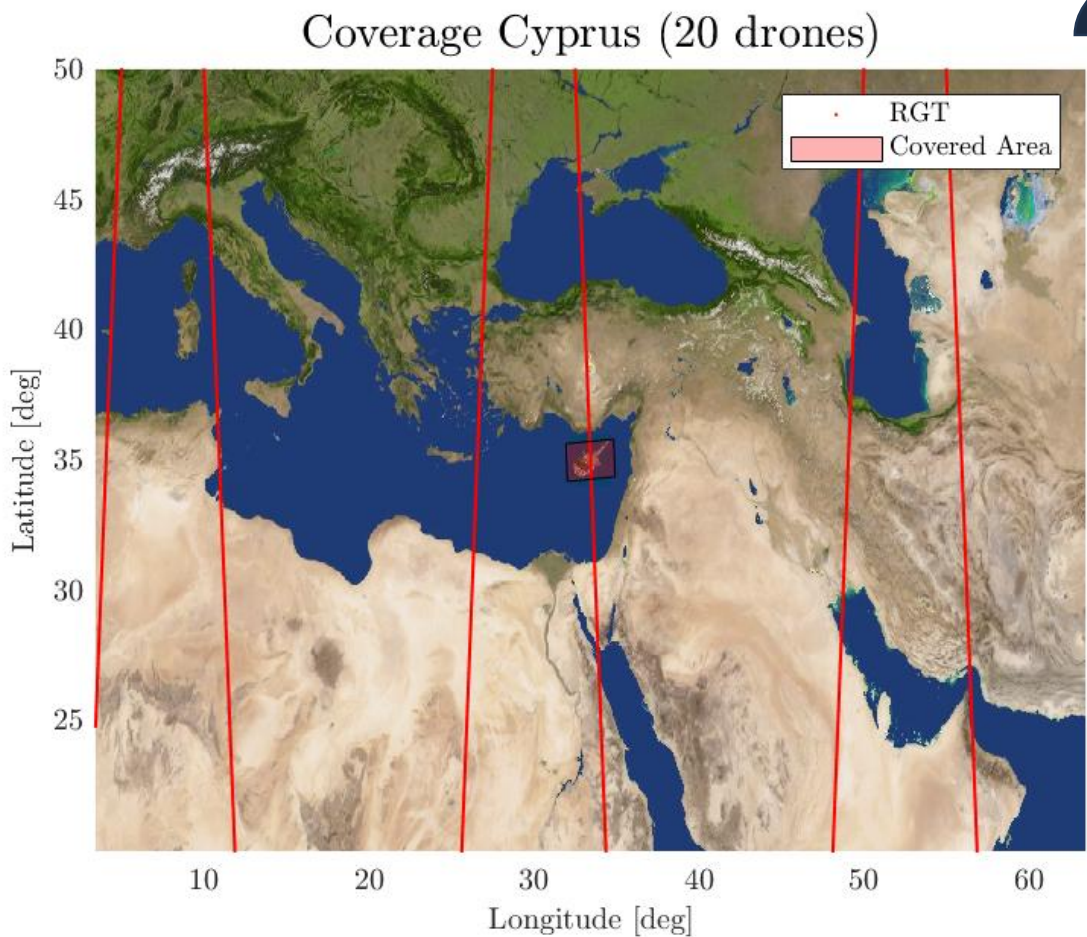
J2 induces a drift which makes it impossible to guarantee consistent visibility over the target for multiple days.

Latitude [deg]	0	15	30	60
Days	8.35	8.64	9.34	16.69

Observation window in days for different latitudes for a single satellite with an inclination of 88°



Single drone in **polar orbit** to have $\Delta\Omega = 0$

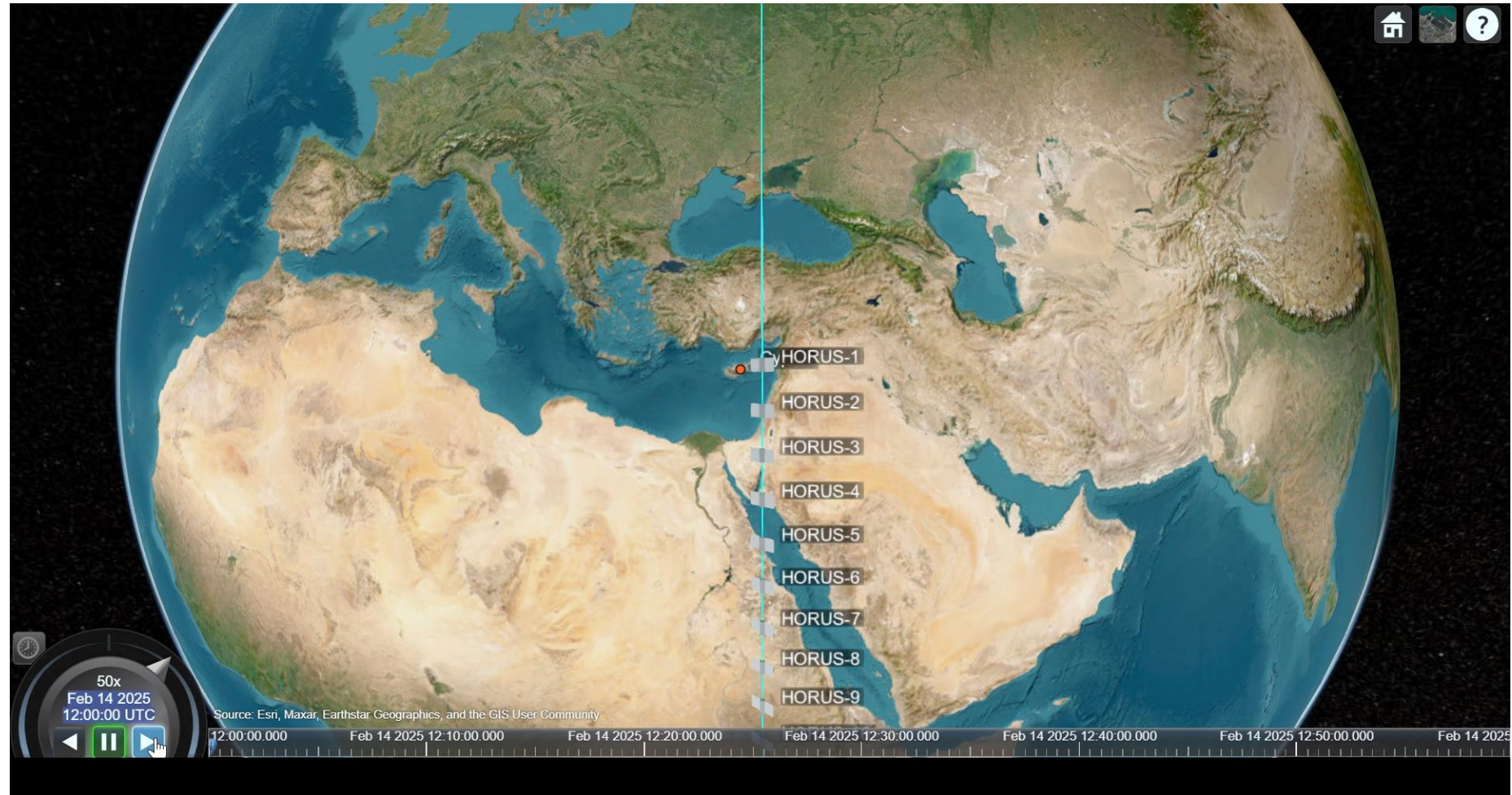


Drivers:

- Coverage area **flexibility**
- Minimize **launch costs**
- $GSD < 1\text{ m}$

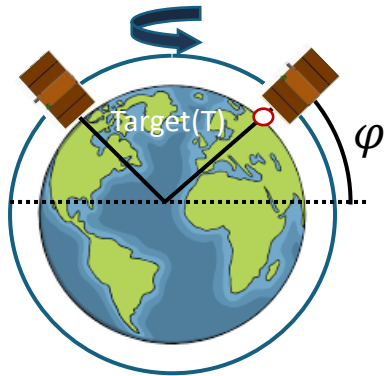


Formation of drones
lined up in the same
orbital plane



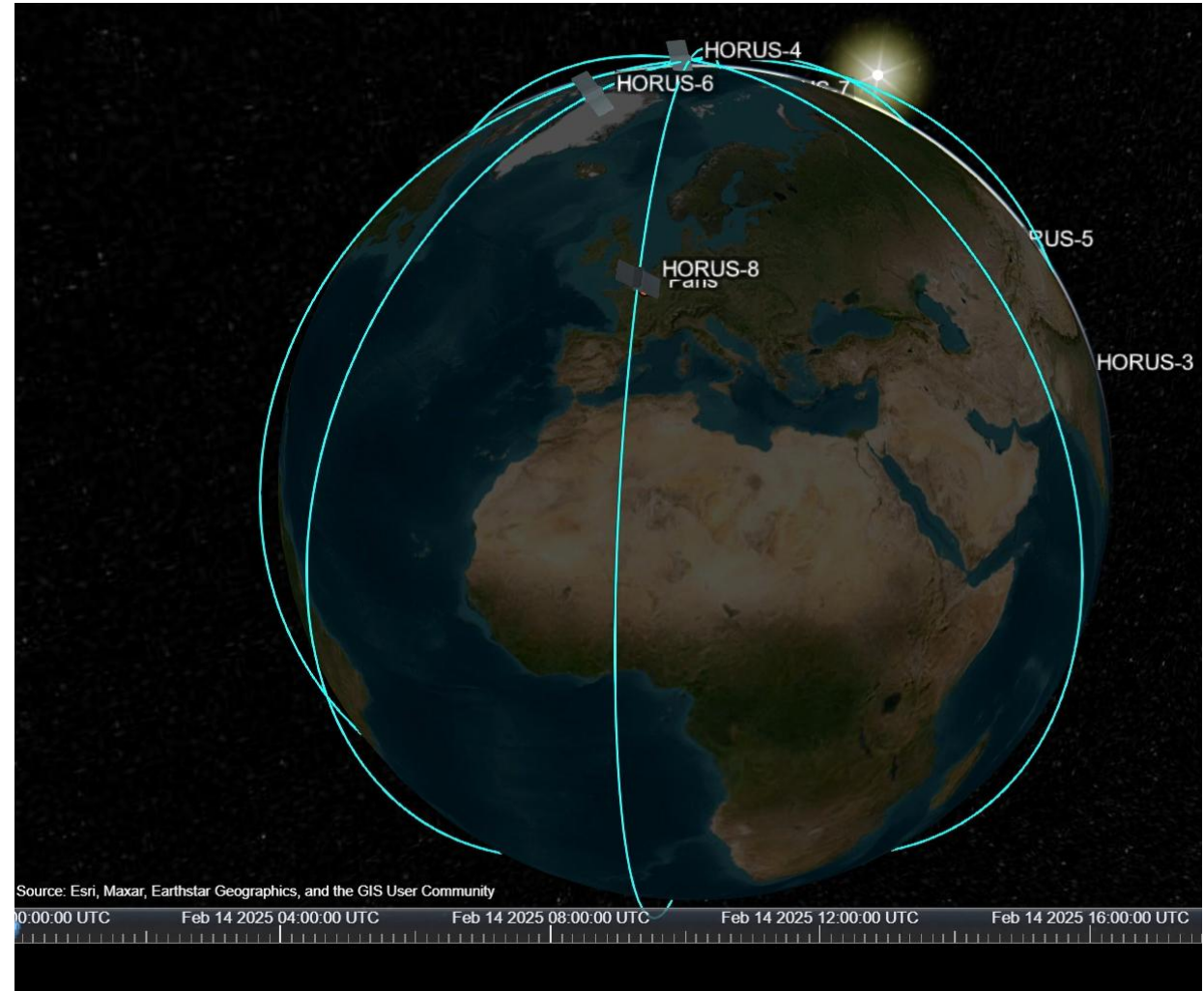
SCALABLE SOLUTION PHILOSOPHY

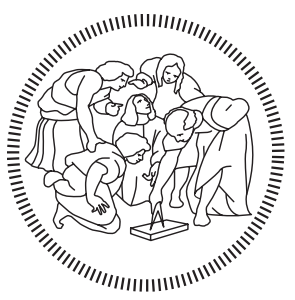
Double the revisit time exploiting the same orbital plane depending on **target latitude**



Further improvement by adding **several orbital planes**

THE CUSTOMER CAN SELECT **ANY LOCATION** AND ASK FOR **ANY TIME SCHEDULE**





POLITECNICO
MILANO 1863

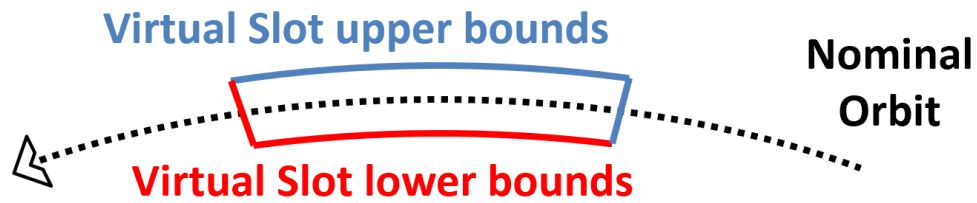


Guidance Navigation and Control

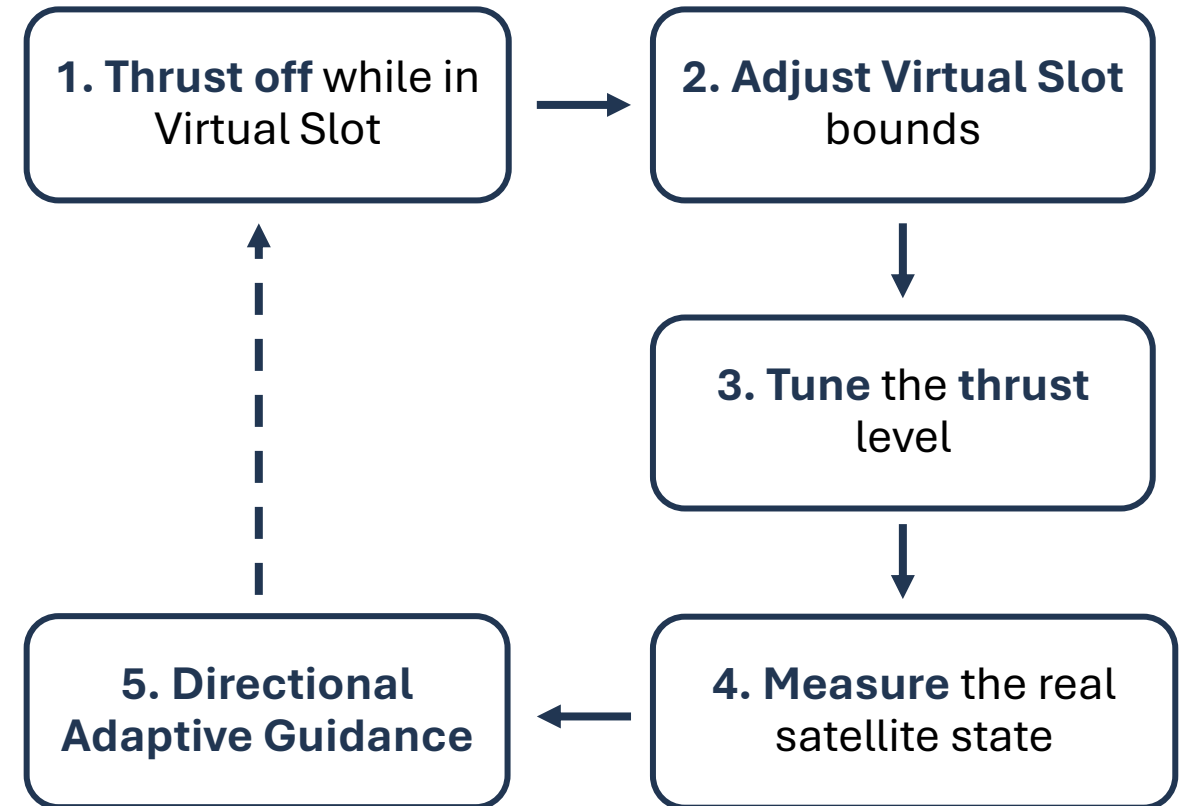
School of Industrial and Information Engineering
Master of Science – Space Engineering

Station Keeping Strategy

1. **Virtual Slot** definition
2. **Thrust** level and slot bounds **tuning**
3. **D**irectional **A**daptive **G**uidance

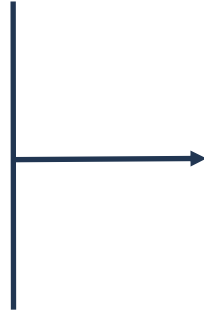


Station Keeping Workflow



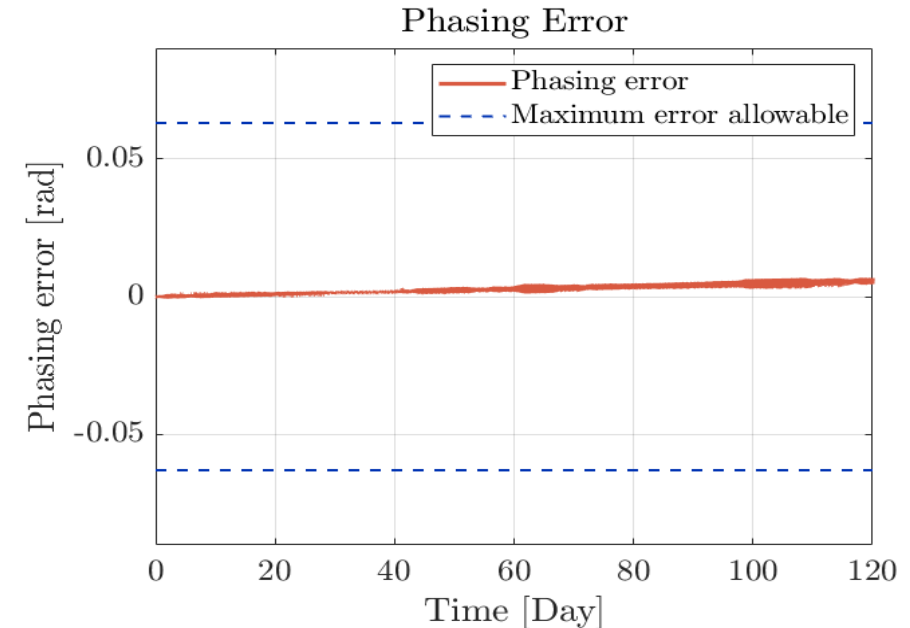
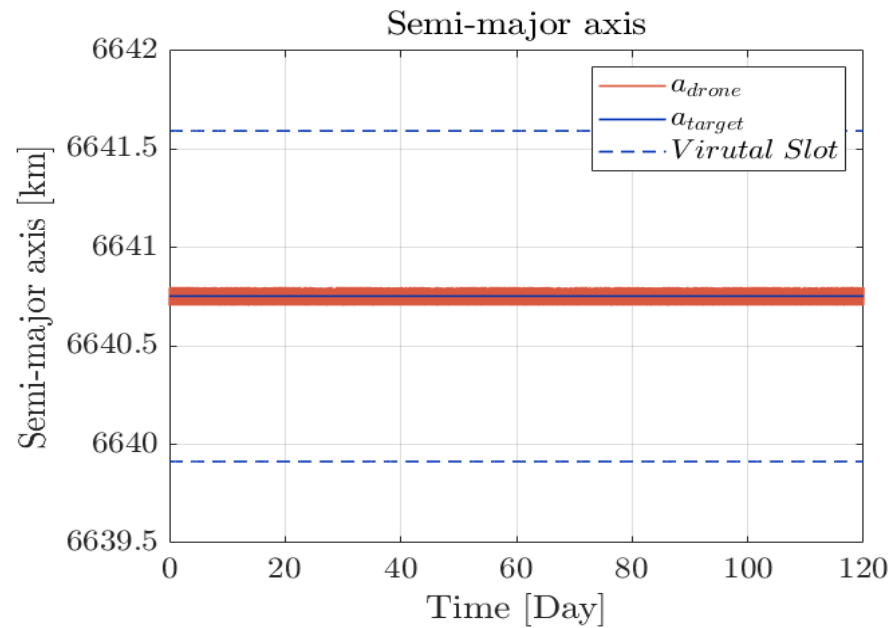
Simulation^[1]

- Maximum **Solar activity**
- Minimum ballistic coefficient
- **Long-term** propagation

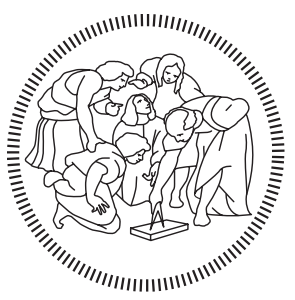


Conclusions

- **Semi-major axis** and **phasing** error are consistently maintained within target bounds
- Additional **propellant margin** available for **non-nominal** operations



[1] A.J. Pacursa et al., "High-Resolution Imaging in VLEO: Preliminary Mission Analysis and GNC Design for the HORUS Mission", IAC-25,C1,IPB,1,x102631. **02 Oct. Screen 31 (2.40 p.m)**



POLITECNICO
MILANO 1863



Tracking, Telemetry and Telecommand

School of Industrial and Information Engineering
Master of Science – Space Engineering

Challenges

Data shall be **downloaded** within **1 h after acquisition**



Coverage analysis to ensure requirements are met from every possible starting point worldwide

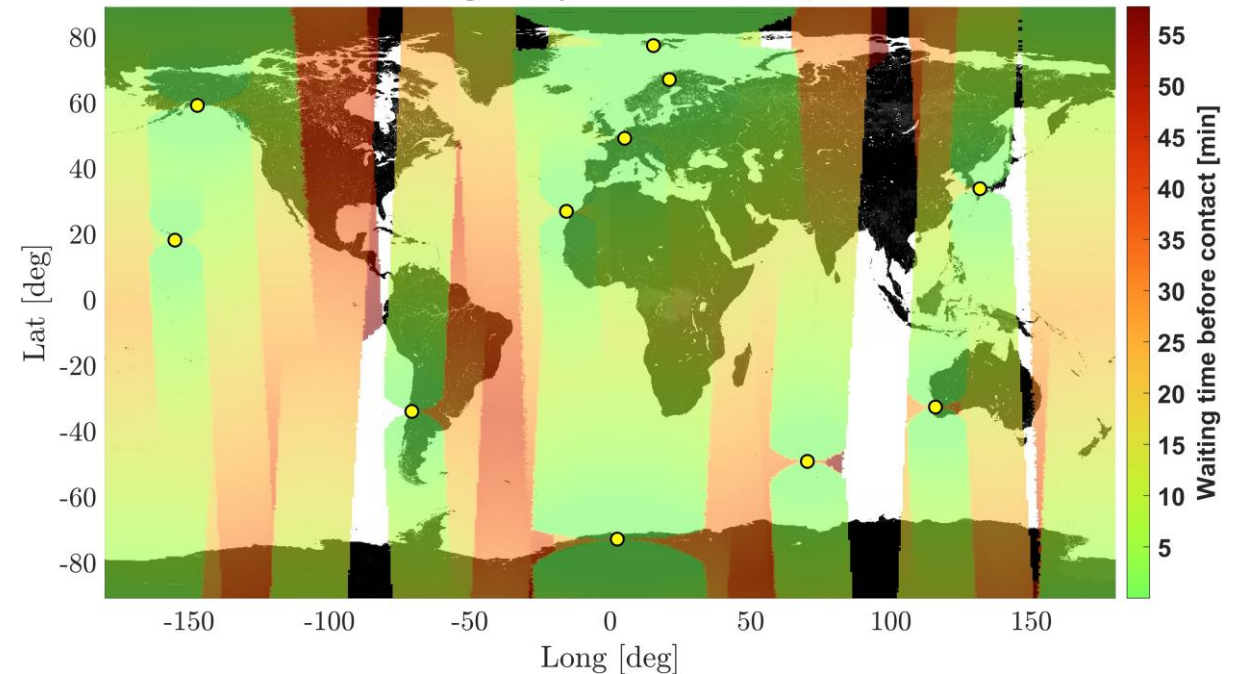
First Iteration

- Only **ESTRACK** ground network included
- Supposing to have target in any latitude of the globe
- HORUS requires at least **53 seconds** for full data downlink.



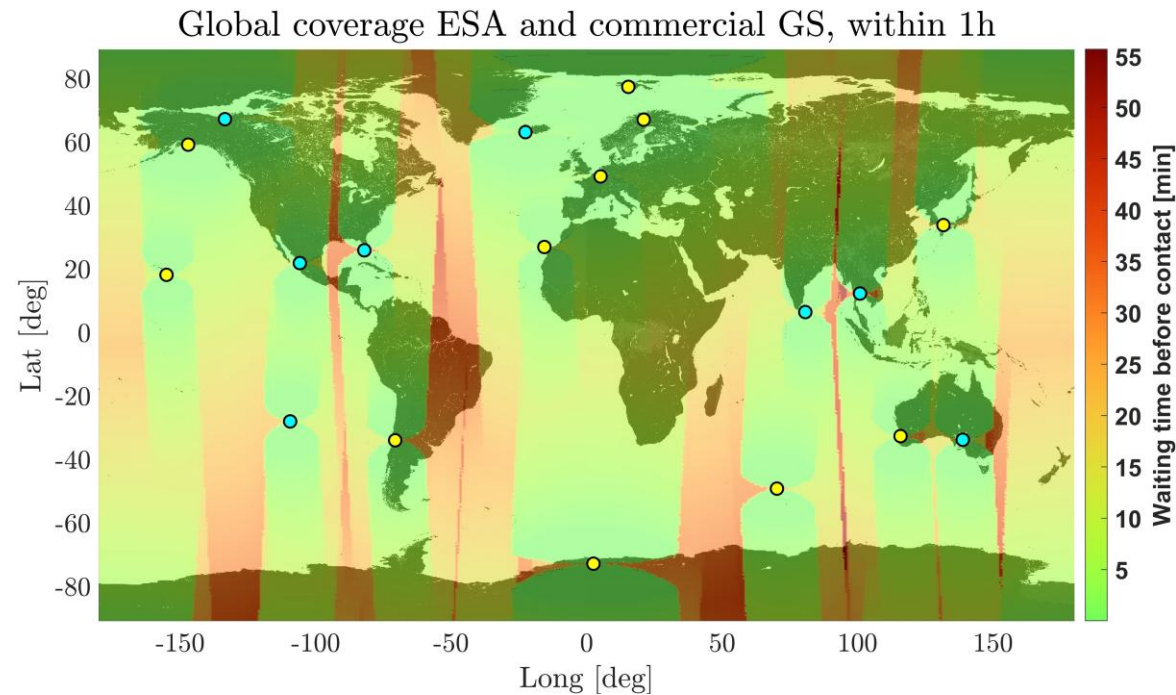
The spacecraft **could not** achieve complete data downlink within one hour post-acquisition from every starting point worldwide

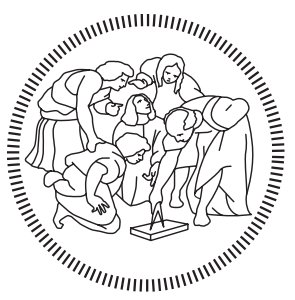
Global coverage only ESA network, within 1h



Second iteration

- The **ESTRACK** network alone **couldn't meet the mission's requirements**
- Ground stations from European **private companies** added to the network to improve coverage and reduce latency.
- The requirement is satisfied when starting from any point on the globe



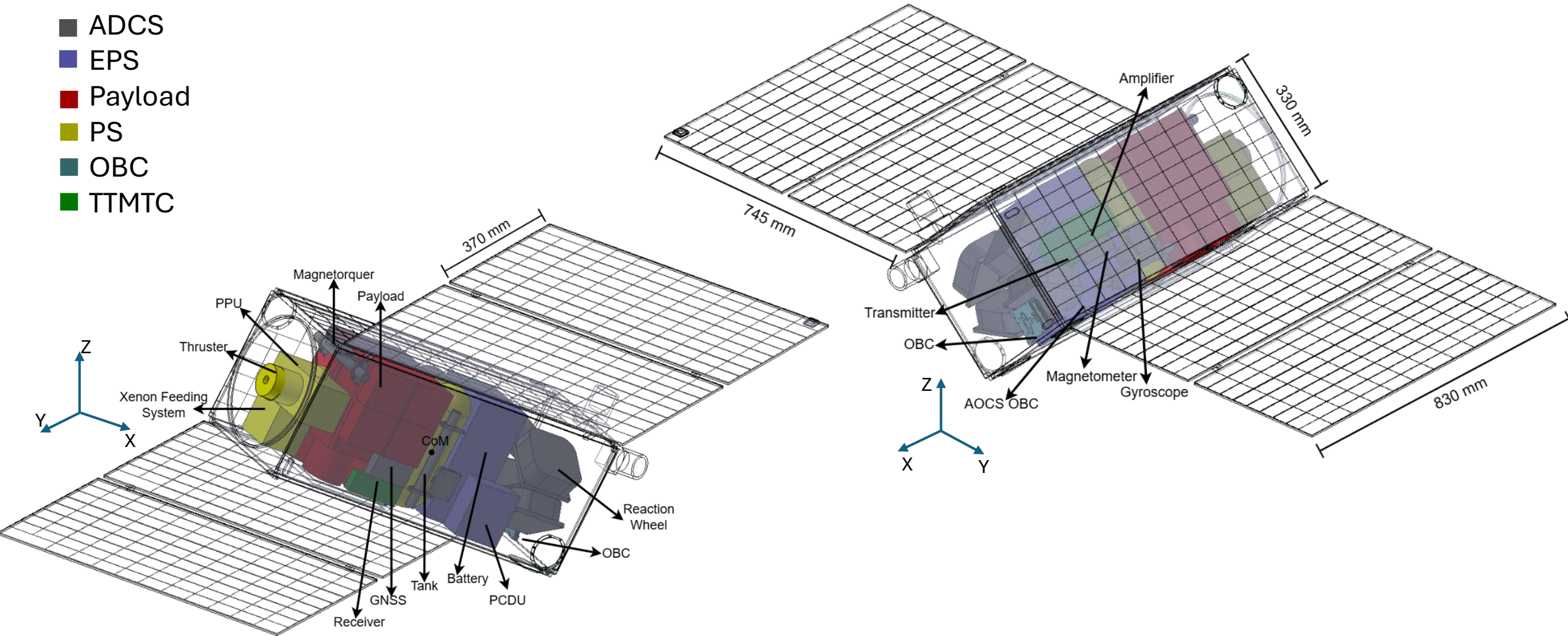


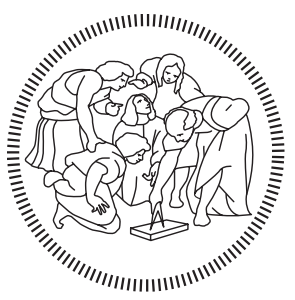
POLITECNICO
MILANO 1863



OTHER SUBSYSTEMS

- ADCS
- EPS
- Payload
- PS
- OBC
- TTMTTC





POLITECNICO
MILANO 1863



Cost Analysis

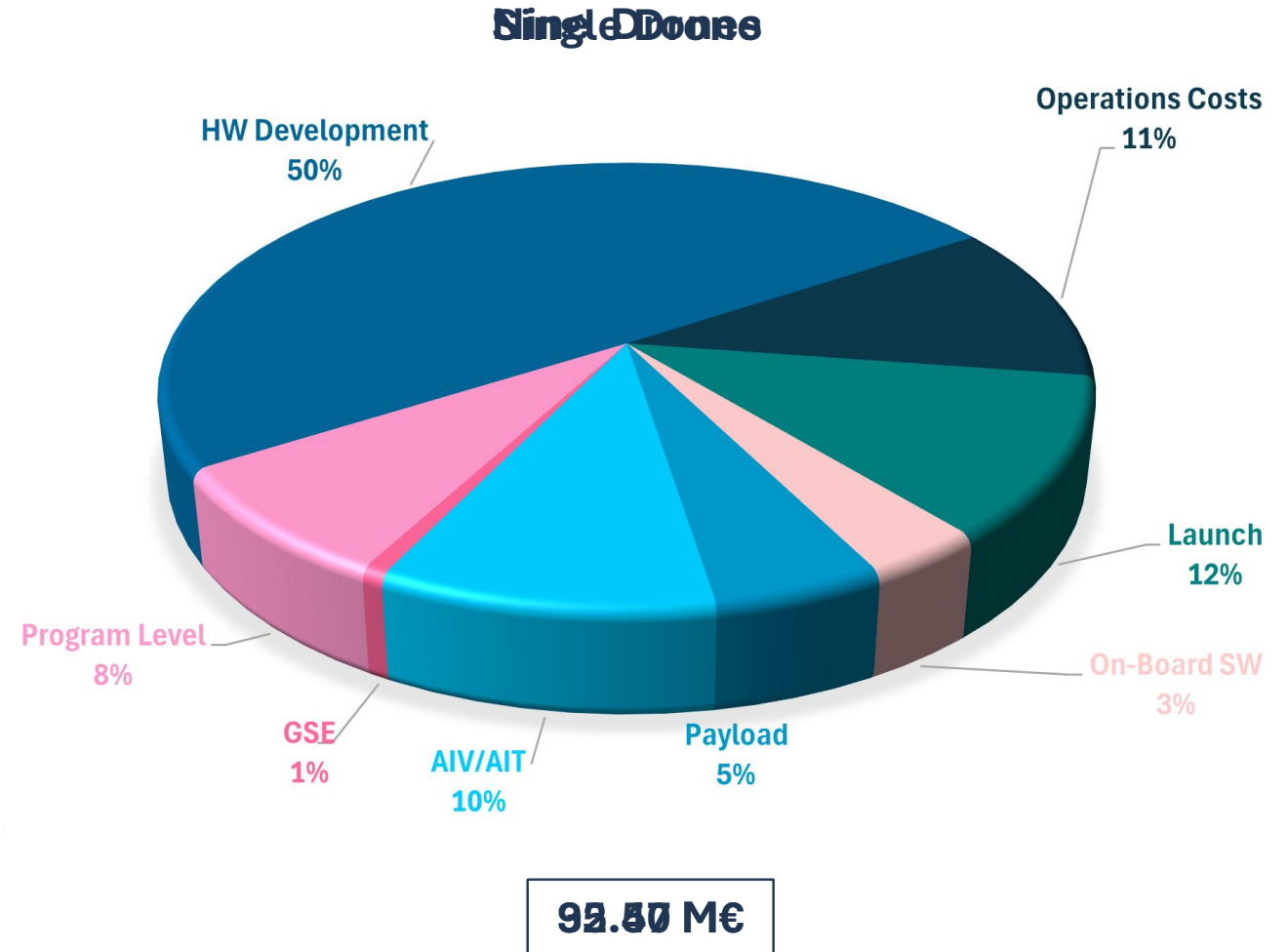
School of Industrial and Information Engineering
Master of Science – Space Engineering

Space Segment

- **Parametric approach** based on CERs
- HW components are all **COTs** except for tanks, ADCS software and TCS heat switchers

Launch & Operations Costs

- **Vega-C** launcher, with a maximum of **9** drones.
- Estimated duration for operations of **one** year.



Final Considerations

- The initial HORUS design successfully demonstrated the **mission's feasibility**.
- Developed a **highly flexible drone** with diverse potential applications
- HORUS could be utilized for: Environmental monitoring, Real-time disaster response, Dual-use operations, Maritime surveillance, Custom campaigns

Future Developments

- Refine and enhance the design by exploring new technological advancements, such as air-breathing engines or air-born launchers
- The **long term goal** is to make HORUS accessible to institutions, NGOs, and commercial users for customized scientific, humanitarian, and commercial campaigns



POLITECNICO
MILANO 1863

Thank you for your attention



Bellini Davide



Beretta Beatrice



Biraghi Giacomo



Braga Matteo



Cappuccio Manuel



Cataffo Francesco



Cesarini Lorenzo



Dowell Matteo
David



Juvara Matteo
Giovanni



Pacursa Andrea
Jimmy